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3.1 Search and rescue as a safeguard at sea

Search-and-rescue (SAR) at sea is an institutionally constituted safeguard against death from maritime distress. International law places a duty on states and on every shipmaster to render assistance to persons in distress at sea, regardless of nationality or legal status (Moreno-Lax, 2018). In the Central Mediterranean this duty is operationalised through a dispersed network of actors — state navies and coast guards, EU agencies, international organisations, and a substantial civil-society fleet of NGO rescue vessels — whose legal mandates differ but whose shared output, when the system functions, is the recovery of vessels in distress and the prevention of drowning. NGOs reference "the international conventions of the sea, [...] the Refugee Convention and the human rights conventions"; state agencies reference national frameworks; coast guards in non-EU states reference border-control mandates (Düvell & Denaro, 2024). The institutional response to a vessel in distress, then, is what stands between physical hazard at sea and death. When it functions, rough seas can be recovered into postponed crossings; when it does not, the same rough seas has a high probability of translating into deaths.

$P(\text{death} \mid \text{rough seas, no SAR})$ is high

This thesis takes that institutional response — operationalised below as the slope of expected deaths with respect to wave height — as its object of study. We refer to the underlying argument as the *guardrail* view of SAR: like guardrails on a highway, SAR does not eliminate the hazard (the rough seas, the sharp curves), but it limits how the hazard becomes harm.

3.2 The 2017 shift: from rescue to interception

Before 2017, the Central Mediterranean was patrolled by a substantial European rescue capacity in international waters. The Italian state-led Mare Nostrum operation (2013–2014), the EUNAVFOR MED Sophia mission, and a fleet of NGO vessels operating close to the Libyan coast — Cusumano & Villa (2021) documents that NGOs alone assisted approximately 120,000 migrants between 2014 and 2019 — together constituted the rescue infrastructure of the corridor. This infrastructure was politically contested but operationally substantial.

The 2017 Italy–Libya Memorandum of Understanding (MoU) marked a qualitative shift to what Düvell & Denaro (2024)'s interviewees describe as "a pull-back regime." The Minniti–Gentiloni Code of Conduct restricted NGO operations, and subsequent Salvini decrees criminalised NGO activity during a period in which "NGOs were persecuted" (Cusumano & Bell, 2021; Cusumano & Villa, 2021; Düvell & Denaro, 2024). The Italian Maritime Rescue Coordination Centre is reported to have withdrawn from international waters (Düvell & Denaro, 2024); EUNAVFOR MED Sophia was downscaled and replaced by Operation Irini, primarily an arms-embargo rather than a rescue mission. In parallel, EU funding built up the operational capacity of the Libyan Coast Guard (LCG), whose mandate is to return intercepted migrants to Libya rather than to disembark them in safety. The post-2017 rise in LCG interceptions is documented in the Frontex and IOM data we use in § [Methods]. Returnees face well-documented violence at the point of interception and abuse in Libyan detention facilities; Düvell & Denaro (2024) summarises UN Human Rights Council findings that migrants across Libya are "victims of crimes against humanity."

Moreno-Lax (2018) names this transition the consolidation of a "rescue-through-interdiction / rescue-without-protection" paradigm: the language of rescue is retained but its operational content shifts from disembarkation in safety to return to a place of harm. The shift is qualitative, not just quantitative. Where pre-2017 rescue was conducted predominantly by actors operating under maritime and humanitarian law, post-2017 the dominant institutional actor in international waters off Libya is one whose mandate is incompatible with the duty to deliver rescued persons to a place of safety. The reduction in SAR is therefore also a reduction in the *kind* of safeguard — not only a reduction in its quantity.

3.3 Why measuring SAR's effect on mortality directly is hard

A direct test of "did SAR removal cause more deaths" would require, at minimum: (a) a credible counterfactual for what SAR exposure would have been absent the MoU, (b) a complete count of deaths under each regime, and (c) a complete count of attempted crossings against which to compute a fatality rate. None of these are cleanly available. The data limitations are not minor: they are structural to the setting.

Departures are not observed. No public dataset records all departures from the North African shore. Frontex records events its operational assets engage with; NGOs record their own incidents; smugglers keep no records that reach analysts. The most reliable approximation of attempted crossings — arrivals to Europe plus pullbacks plus deaths — is itself constructed from the partial records of each actor and is bounded below by what each actor reports. Without a credible denominator, fatality *rates* cannot be computed without strong assumptions, and any "deaths per crossing" comparison inherits the bias of every component dataset.

Recorded deaths are a selected subset of true deaths. The empirical analysis therefore treats UNITED as the primary mortality source and uses IOM Missing Migrants Project estimates as a robustness comparison. This choice is substantive rather than cosmetic: the two projects use different reporting and verification procedures, and in the corridor sample used here UNITED records more deaths than IOM. IOM remains useful because it is widely used and transparently documented, but it is more closely tied to located, witnessed, or otherwise reported incidents. Boats that sink unwitnessed may be absent from either source. In the data used in this thesis, fewer than half of IOM-recorded death events have a corresponding Frontex incident on the same date and zone — direct evidence that a substantial share of fatal events occurs outside the European operational gaze. Düvell & Denaro (2024) describes the rescue and surveillance network of the corridor as "vast" and "patchy" and reports that "the vast space and the patchy surveillance and rescue network, as well as the continuously high level of lives lost at sea, demonstrate that rescue remains unreliable, that migrants and refugees cannot count on rescue and that the risk of dying at sea is significant." The unreliability of rescue is also the unreliability of the recording of death.

The selection mechanism for what gets recorded changes when the policy regime changes. Before 2017, NGO presence was a major source of incident documentation; after 2017, NGO retreat reduces the rate at which incidents in international waters are seen and reported. From 2019, the Italian coastguard reclassified most SAR operations as "law enforcement" operations, which Düvell & Denaro (2024) reports introduces ambiguity into the official statistics. Pullbacks remove migrants from the at-sea population without generating mortality records: a returnee may face detention, abuse, or death in Libya, but not in the IOM mortality dataset. The upshot is that *whatever bias exists in observed death counts is likely to have shifted in magnitude, and possibly in direction, around the policy event.* A naive comparison of pre-MoU and post-MoU recorded deaths therefore conflates genuine change in mortality with change in mortality measurement. Plausibly, post-MoU under-recording is *worse* (fewer rescue assets to witness sinkings), which would make naive estimates *understate* the true mortality change.

These three problems compound. A naive "deaths per crossing, before vs. after the MoU" estimate would have an incompletely measured numerator divided by an unobserved denominator, with both subject to selection mechanisms that shifted at the policy event. The strategy this thesis adopts therefore treats the daily UNITED death count as the primary reduced-form outcome, estimates the same models on IOM as a comparison, and reports constructed-rate models only as risk-oriented sensitivity checks.

3.4 The wave–mortality slope as a sensible empirical strategy

The estimand we adopt is how the *slope* of recorded mortality with respect to lagged daily mean significant wave height (SWH) changes across SAR/policy regimes. The main empirical implementation models daily recorded deaths, using UNITED as the primary series and IOM as an independent comparison. A Poisson offset specification divides deaths by a source-specific constructed-attempt denominator, but this is secondary: the denominator is not observed daily departures, and the offset imposes a unit elasticity between attempts and deaths that the data need not satisfy. Three properties make the slope estimand informative under the data limitations described above.

First, **SWH is exogenous to the policy regime**. Atmospheric and oceanic conditions do not respond to the Italy–Libya MoU. The same physical hazard — a 2-metre wave on a given day — acts on a small overloaded vessel in the same physical way regardless of whether NGO ships are at sea, regardless of who is patrolling, regardless of which mandate the patrolling actor operates under.

Second, **the rate model is informative but weaker than the count model as a primary design**. Daily deaths change both because crossings become more or less lethal and because the number of people attempting the crossing changes. A risk estimand would therefore require daily departures, but those departures are unobserved. The rate specification uses a constructed lower-bound denominator: Frontex-recorded persons plus LCG/TCG pushbacks plus deaths from the same outcome source. For IOM outcomes the offset is based on IOM deaths; for UNITED outcomes the offset is based on UNITED deaths. This source-specific construction avoids putting IOM deaths in the UNITED denominator, but it remains an approximation rather than observed exposure and is reported as sensitivity evidence.

Third, **the slope on SWH is robust to many natural patterns of under-reporting in death data**. If under-reporting is approximately multiplicative — some constant fraction of true deaths is missed, regardless of weather — then under-reporting affects the *level* of recorded deaths but leaves the *slope* on weather essentially unchanged. The claim "the slope shifted around the MoU" is then identified even if the death series is biased downwards in both regimes. The remaining threat is *differential* bias: if under-reporting changed differently with SWH across regimes, the slope estimate would mix genuine change with reporting change. We discuss this concern explicitly in [Methods] and probe it by replicating the analysis on an independently sourced death series ([Results]).

This logic mirrors the methodological move of Deiana et al. (2024), who use day-to-day SWH variation as an exogenous source of identifying variation in the same corridor. The estimand they target is the slope of *crossings* on SWH; the main estimand here is the slope of *recorded deaths* on SWH, with constructed-rate estimates reported as a secondary risk-oriented check. Both inherit the exogeneity of the underlying weather variation; both are silent about parts of the system that cannot be observed.

3.5 Competing predictions

The four readings of SAR in the literature imply distinct predictions for the SWH–mortality slope.

The *pull-factor* view holds that SAR primarily affects the volume of attempts and is broadly silent on per-crossing risk; the slope between waves and deaths should be roughly invariant to SAR regimes (Rodríguez Sánchez et al., 2023). The *moral-hazard* view holds that SAR availability shifts the composition of crossings toward unsafe vessels, which become more weather-sensitive; the slope should *steepen* when SAR is *abundant* (Deiana et al., 2024). The *deterrence-and-diversion* view holds that the post-2017 regime deterred crossings on the CMR and diverted some flow elsewhere; falling mortality after the MoU could reflect a falling denominator rather than safer crossings, and the slope on weather need not change (Camarena et al., 2020; Hoffmann Pham & Komiyama, 2024; Tafani & Riccaboni, 2025). The *guardrail* view, which this thesis develops, holds that SAR primarily affects the per-crossing fatality rate by determining whether distress is resolved; the slope should *flatten* when SAR is abundant and *steepen* when SAR is removed.

The closest precedent on the same policy event is Zambiasi & Albarosa (2025), who estimate a spatial difference-in-differences on the 2017 MoU and find an increase in the probability of a deadly incident in areas patrolled by the LCG after the deal. Their result is consistent with the guardrail prediction in the spatial dimension; this thesis pursues the temporal dimension — the change in the SWH–mortality slope — which Zambiasi & Albarosa (2025) do not estimate.

The four views are not mutually exclusive: SAR could simultaneously shift volume, composition, per-crossing risk, and route choice. They imply distinct empirical signatures, however, and the SWH–mortality slope is the signature most directly tied to the guardrail view.

3.6 Theoretical estimand

Let $M(s, R)$ denote expected recorded deaths per constructed crossing attempt on a day with significant wave height s and SAR capacity R. The guardrail view predicts that the partial derivative of M with respect to s is more negative (or less positive) when R is high than when R is low. The theoretical estimand is the difference between these two slopes:

$$\Delta\beta \;=\; \left.\frac{\partial M(s,R)}{\partial s}\right|_{R=R_{\text{low}}} \;-\; \left.\frac{\partial M(s,R)}{\partial s}\right|_{R=R_{\text{high}}}$$

A positive $\Delta\beta$ means that the same rough day is associated with a higher recorded death rate when SAR has been removed than when SAR is in place. This is the guardrail prediction stated as a slope comparison, not as an estimate of lives saved by SAR.

3.7 From theoretical to empirical estimand

We do not observe SAR capacity as an exogenous variable. We instead observe a policy event that approximately marks a shift between the two regimes described in §1: the 2017 Italy–Libya MoU, in force from 1 July 2017. We treat the MoU date as an indicator of this regime change and define the empirical estimand as the change in the SWH–recorded mortality-rate slope around it:

$$\beta_3 \;=\; \beta_{\text{post}} \;-\; \beta_{\text{pre}}$$

where β_{pre} and β_{post} are the slopes of the conditional expected recorded death rate on lagged five-day mean SWH, before and after 1 July 2017. The count specification estimates the same slope for daily recorded deaths and is reported as supporting reduced-form evidence.

The link from β_3 to $\Delta\beta$ rests on the assumption that the MoU is a meaningful proxy for the SAR regime change of interest. This assumption faces two threats. First, the MoU bundles SAR removal with other contemporaneous changes: the criminalisation of NGOs (Cusumano & Villa, 2021), the Salvini decrees of mid-2018, the broader externalisation logic of EU migration policy (Wolff, 2024), and a documented shift in boat composition (Deiana et al., 2024). Whatever shift in β_3 we estimate is a reduced-form moderation by the policy-regime bundle, not an isolated effect of SAR alone. Second, the policy effect could spread gradually over calendar time, making the MoU date a coarse signal of an ongoing transition.

We address the first threat with a complementary mechanism specification ([Methods]) that replaces the binary policy-event indicator with a continuous measure of SAR capacity — the weekly number of persons rescued by Frontex-coordinated operations — and asks whether the slope tracks SAR capacity directly. We address the second by reporting regime and rolling-gradient decompositions ([Results]).

The remaining gap between the theoretical estimand $\Delta\beta$ and the empirical estimand β_3 is real and we do not paper over it. We estimate how the marginal association between sea state and recorded deaths changes across SAR/policy regimes. Under the assumption that within-month SWH variation is exogenous and no other post-2017 factor changed the weather-to-death gradient, the estimates can be interpreted as policy-regime moderation. That is a strong assumption.

3.8 Research question

This framing yields a single empirical question:

> How does the marginal association between sea state and recorded deaths change across SAR/policy regimes?

A positive shift is consistent with — though not unique to — the guardrail view. A null shift is consistent with the pull-factor view. A shift opposite in sign would be consistent with the moral-hazard view. Distinguishing among these, given the data limitations described in §3, is the empirical task that follows.

4 Methods and Data

5 Analysis and Results

6 Conclusion and Further Work

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Appendix